

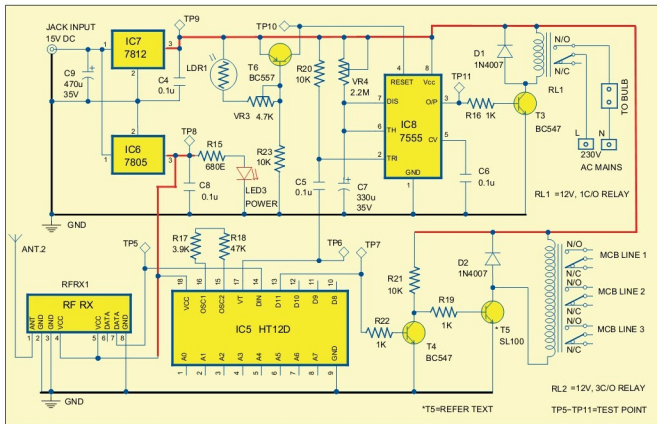


Fig. 3: Receiver circuit

HT12E encoder. It is an 18-pin DIP encoder IC whose operating voltage is 2.4V to 12V. It is capable of encoding information consisting of 'N' address bits and 12-N data bits. The programmed addresses/data are transmit-

ted together with header bits via an RF or infrared transmission medium upon receipt of a trigger signal.

RF-TX. It is a 4-pin ASK RF transmitter module operating at a frequency of 433 MHz. It requires 1.5-12V DC to

operate. The data input is fed to its pin 2 from encoder and the signal is transmitted from its pin 4 connected to an antenna (ANT.1). The antenna can be a simple hook-up wire of about 10cm length.

Receiver. The receiver section consists of RF-RX receiver, HT12D decoder, LDR sensor, power supplies, relay drivers and relays for switching on/off the electrical appliances and mains power line.

RF-RX. It is a miniature ASK receiver module that operates off 5V DC. It receives on-off keyed (OOK) modulation signal and demodulates it to digital signal. It is based on single-conversion, super-heterodyne receiver architecture and incorporates phase-locked loop (PLL) for precise local oscillation generation. The RX-TX is

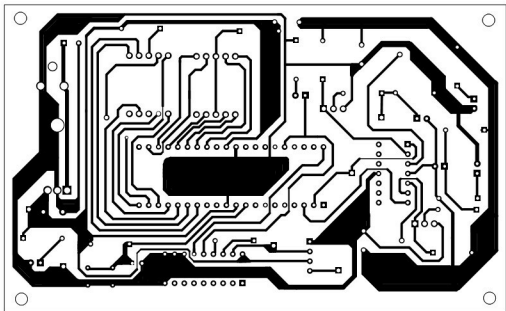


Fig. 4: Actual-size, single-side PCB for the transmitter